

Cover letter — resubmission

To: Dr. Lina Ma, Editor-in-Chief, and the Editorial Office
Bioinformatics Advances

Dear Dr. Ma and Editors,

Please find enclosed an **Original Article**, *“Leakage tracks how a benchmark was assembled, not what it contains: a curation defect in genomic sequence benchmarks and the model rankings it inverts.”* This is a resubmission of **BIOADV-2026-296** (previously *“Homology leakage selectively changes which model wins on genomic sequence-classification benchmarks,”* Associate Editor: Prof. Shanfeng Zhu), which received a Reject & Revise decision. As instructed it is submitted as new work, with the prior manuscript number given here so the records can be linked. A point-by-point response to every reviewer and Associate-Editor comment accompanies it, together with an 11-page supplement.

The decision turned on whether the ranking result is general or an artifact of one narrow comparison. Three things changed most.

The claim moved upstream, and the paper’s spine changed with it. The prior submission argued that a curation defect caused the leakage; it did not test that. The defect is now located in the shipped genomic coordinates, before any model is trained — 77,421 positive intervals on 40,934 distinct coordinates — and a coordinate screen with no fitted parameter separates seven of the suite’s eight datasets. That construction step is then edited in both directions: applying the omitted merge step to the leaky dataset collapses its leak fraction $0.390 \rightarrow 0.012$ and the random forest’s inflation $+0.165 \rightarrow -0.008$, abolishing the reordering; imposing the same defect on a clean dataset, at matched size and balance with model code and sequences untouched, manufactures it on demand. All four arms carry cluster-bootstrap intervals, and control and intervention intervals are disjoint in both directions. The title now states the resulting claim rather than the symptom.

A pre-registered prediction failed, and the failure is the stronger result. Committed before the target suite was examined, it held that GUE’s short fixed-length human regulatory tasks would be leaky. At full scale all eleven are clean. That refutes the natural reading of our own finding — that short human regulatory DNA is inherently at risk — and is why the paper now claims that leakage tracks assembly rather than sequence type. The registered predictions score 3 of 15 and are reported that way.

The deep-model and multiclass requests are answered rather than deferred. A from-scratch 1D CNN, seed-replicated, ranks fifth of ten on the leaky split and first on the corrected one on every one of five seeds (paired difference against the linear SVM $+0.0111$, combined-source CI $[0.0019, 0.0204]$) — uniform in rank though not in significance, since four of the five seeds’ paired intervals exclude zero and the fifth does not. Leakage here does not only promote a memorizing tree ensemble, it demotes the convolutional architecture whose baselines this suite publishes. The prior submission conceded that its multiclass arm was classical-only; that arm has since been run. Under injected leakage on the three-class set, the CNN shows no dose-response across three seeds (mean drops $-0.0386, -0.0182, -0.0249, -0.0089$) while a default forest on the identical construction drops $+0.0004, +0.1177, +0.1791, +0.2581$. That null is the paper’s own “memorization propensity, not capacity” claim being tested and holding: the label-carrying copies are available to both learners, and only one exploits them.

The remaining requests are addressed in the response: MMseqs2 alignment-scored validation and a length-robust containment index, which moves one previously-clean verdict to borderline; a

prevalence stress test; a cluster bootstrap over whole near-duplicate components; a regularization path, together with the experiment the hyperparameter objection really demands — ordinary cross-validation on the as-shipped training set *selects* the memorizing default, and the benchmark then shows the leakage-aware tuner’s model 10.9 points worse as shipped while it is 3.3 points better corrected; and the adoption of *near-duplicate leakage* as the primary term throughout.

Three claims from the prior submission were wrong and are corrected rather than quietly dropped. The assertion that de-duplication over-removes genuine promoter and gene-family signal does not survive measurement: the cluster in question is 420/420 negative class and is a single locus tiled by 251 bp windows at a median 2 bp step — window geometry, not paralogy. The pretraining-overlap confound was asserted at $\approx 100\%$; HyenaDNA designates chromosomes 14 and X as held-out pretraining test chromosomes, both present here, bounding overlap at $\leq 93.4\%$ and $\leq 84.7\%$ — which also identifies the confound-free protocol a future pretrained-model evaluation should use. And the design-effect diagnostic was computed on the original arm when the corrected arm is the clustered one; the corrected figures are 2.80 and 2.58 against a mean-size convention of 1.29 and 1.33, with measured variance inflation of $3.96\times$ and $2.12\times$. Every reported interval remains correct as computed, because the block bootstrap resamples those components rather than modelling them.

What did not work is reported as prominently as what did: a pre-registered dose-response that is not one; the first registered prediction failing outright, with the corrected statistic that would favour it labelled *post hoc* rather than counted; a cross-suite null; a simpler competing rule that matches our diagnostic condition’s 10/10 score using none of its inputs; a from-scratch attention arm whose one completed dataset moves significantly in the contrary direction, on which no claim rests; and one dataset carried as an explicit cautionary partial case rather than a second demonstration.

The study is fully reproducible; code, the report card, the near-duplicate-aware splitter and the certification tool are in the linked repository, and the manuscript states its pinned software environment. The work is original, is not under consideration elsewhere, and all authors approve this submission. We have no competing interests to declare.

We hope the revised manuscript now meets the standard for publication.

Sincerely,
Rikhin Kavuru
Independent Researcher
rikhinkavuru@gmail.com